

The Effect of Education on Smoking Behavior: A Causal Inference Analysis Using NHEFS Data

Lyann Sun

1 Introduction

Cigarette smoking represents a significant public health issue in the United States, with more than 16 million individuals affected with illnesses related to smoking [1]. Moreover, smoking remains a "leading preventable cause of death" in the United States, accounting for roughly 480,000 deaths annually, including those resulting from secondhand smoke exposure [1]. Therefore, it is important to identify the factors that influence smoking behavior.

Evidence has shown that education plays a role in smoking behavior- individuals with a lower education level are more likely to smoke and are less likely to quit than their more educated peers. However, the nature of this relationship- whether this relationship reflects a causal effect of education or instead arises from shared background factors remains a subject of debate. Maralani's research shows that early school environments and family context predict long-term schooling attainment and smoking behavior [2]. In contrast, a more recent twin-pair study found that after controlling for genetic and social factors, the sibling with more years of education demonstrated higher rates of quitting smoking and lower overall likelihood of using cigarettes, providing evidence that education has a causal effect on smoking behavior [3].

While previous studies have established a relationship between education and smoking behavior, these studies differ in their conclusions regarding whether education itself has a causal effect on smoking. To address this, our study builds on this literature by estimating the effect of education on smoking using modern causal inference methods. Specifically, we ask: **after adjusting for demographic, lifestyle and mental health covariates, does attaining a higher level of education (college) have a causal effect on the change of smoking behavior?**

2 Data

The dataset used in this project is obtained from the "causalddata" package in R, which comes from the Epidemiologic Follow-up Study of the National Health and Nutrition Examination Survey (NHANES) [4] [5]. This data tracks participants from an initial NHANES medical examination in 1971-1975 and includes follow-up data collections in 1982-1984, 1986, 1987 and 1992. Data was collected through personal interviews, medical examinations and personal records, providing longitudinal information on health behaviors, socioeconomic factors and health outcomes over time.

Since we are interested in the effect of education on smoking behavior, we used the variable **education**, which indicates the amount of education by 1971, to define the treatment. We created a new binary indicator column **high_edu**, with a value of True if the individual completed at least a college-level education (education category 5), and False otherwise (categories 1-4). The outcome variable **smkintensity82_71** measures the change in the number of cigarettes per day between 1971 and 1982.

In addition, we include potential confounders that can be grouped into three categories: demographic, lifestyle and mental health. The demographic covariates are: **sex** (male or female), **race** (white or otherwise) and **income** (total family income in 1971). The lifestyle covariate is **alcoholfreq**, which measures alcohol frequency consumption in 1971. For mental health, we included the variable **nerves**, which indicates if the individual used nerves medication in 1971. More information about these covariates can be found on Appendix A.

After removing any rows with nulls, the total number of rows used in this study is 1,502, in which 172 are treated and 1,330 are controlled as seen on Figure 1. Please refer to Appendix A and B for summary statistics and a sample of the data.

3 Potential Outcomes

Our goal is to determine the causal effect of having at least a college-level education on subsequent smoking behavior. The treatment is defined as having completed a level of education of at least college by 1971, and the control is having a level of education less than college by 1971. The outcome is the change in cigarettes smoked per day between 1971 and 1982.

Hence, the potential outcomes are:

- $Y_i(1)$: the change in the number of cigarettes smoked per day between 1971 and 1982 if individual i had completed at least a college-level education by 1971.
- $Y_i(0)$: change in number of cigarettes smoked per day in 1971 and 1982 if individual i has completed less than a college-level education by 1971.

Our targeted estimand is the average treatment effect (ATE), defined as:

$$E[Y_i(1) - Y_i(0)]$$

Per SUTVA, there should be no hidden versions of the treatment and no interference across individuals. The no hidden version of the treatment assumption could be violated if the individuals experience different forms of college education. For example, one person could have graduated from online community college courses, while another person could have graduated from a 4-year residential college institution.

The no interference assumption could be violated if individuals that have a level of education less than college are influenced by peers who have a higher level of education. For example, if a college-educated friend discourages a person without a college degree from smoking, this could affect this person's smoking behavior and thus their outcome.

4 Methods and Results

To estimate the causal effect of education on smoking intensity, we employed two methods: **pair matching** and **outcome regression**. Both methods aim to estimate the effect of having at least a college-level education by 1971 on subsequent changes in smoking behavior.

4.1 Identification Assumptions

For identification of the causal effect, we rely on three standard assumptions: SUTVA, conditional ignorability, and conditional overlap.

First, SUTVA is assumed to hold as discussed in the previous section. Specifically, we assume no hidden version of treatment and no interference between units and treatments as well as no hidden versions of the treatment.

Second, for conditional ignorability, we assess the plausibility of the assumption using a causal diagram(Figure 2.a). Based on the prior literature on this topic, we identify three main categories of potential confounders: demographic factors, lifestyle factors and mental health factors. Conditioning on these variables is assumed to block all backdoor paths between treatment Z and outcome Y , thereby rendering treatment assignment independent of potential outcomes given covariates, namely, strong ignorability holds:

$$(Y(1), Y(0)) \perp Z | X$$

This assumption may fail if there exist unmeasured confounders, such as cognitive ability, which influence both education and smoking behavior.

Finally, for the conditional overlap assumption, we evaluate the distribution of estimated propensity scores between treated and control groups. The propensity score is estimated by logistic regression and defined as:

$$e(X) = P(Z = 1 | X)$$

where $Z = \text{high_edu}$, $X = (\text{age, sex, race, income, alcoholfreq, nerves})$.

The empirical distributions (Figure 2.b) show substantial overlap in their support, indicating that for each level of covariates, there is a positive probability of receiving both treatment and control. Therefore, the conditional overlap assumption is considered to be reasonably satisfied. This assumption may fail in certain group of people (very high income or very low income individuals) almost deterministically receive one treatment level.

4.2 Pair Matching

We first apply pair matching to estimate the causal effect. For the pair matching approach, each treated subject (individuals with at least a college education) was matched to a control subject(individuals with less than a college education). Matching was performed by minimizing the absolute difference in the estimated logit propensity scores between matched pairs. The propensity score model included pre-treatment covariates including **age, sex, income, nerves medication use, drinking frequency, race** altogether 6 variables. This

procedure aims to improve covariate balance between treatment and control groups, as illustrated by the Love plot(Figure 3), which shows great improvement in covariate balance after matching.

After matching, 172 pairs were generated. We assessed the treatment effect by comparing the change in smoking intensity between 1971 and 1982 within matched pairs. Specifically, we tested the sharp null hypothesis of no treatment effect using a paired Wilcoxon signed rank test.

Let Y_i^T and Y_i^C denote the outcomes for treated, matched control units, the estimated mean difference in the outcome(treated minus control) was:

$$\hat{\tau}_{matching} = \frac{1}{n} \sum_{i=1}^n (Y_i^T - Y_i^C) = -3.53$$

indicating that individuals with higher education experienced, on average, a larger reduction (or smaller increase) in smoking intensity compared to their matched counterparts with lower education.

Moreover, the Wilcoxon signed rank test (Table 1) provided statistically significant evidence against the null hypothesis with p-value 0.019 and estimated 95% confidence confidence interval:

$$CI : [-6.50, -0.00]$$

Under the assumption of strong ignorability and correct specification of the matching and inference procedure, these results suggest that there is statistically significant evidence that higher education causally affects changes in smoking intensity between 1971 and 1982 in this sample.

4.3 Outcome Regression

We next estimate the causal effect of education on the change in smoking intensity between 1971 and 1982 using outcome regression approach. Note that for this approach, we are assuming linearity of the regression model.

Let $\mu_1(X_i) = E(Y_i|Z_i = 1, X_i)$ and $\mu_0(X_i) = E(Y_i|Z_i = 0, X_i)$, where the covariates we selected includes **sex, income, age, nerves medication use, drinking frequency and race**, the outcome regression estimator is defined as:

$$\hat{\tau}_{reg} = \frac{1}{n} \sum_{i=1}^n \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$$

which targets the average treatment effect(ATE) under correct specification of the outcome models.

We estimate μ_1 and μ_0 using linear regression models, the estimated ATE is:

$$\hat{\tau}_{reg} = -4.01$$

which aligns with the result from matching, indicating that on average, having at least a college-level education is associated with a reduction in the change of smoking intensity

between 1971 and 1982.

For asymptotic variance, we get:

$$\hat{V}_{superpop} = \frac{\sum \epsilon(1)}{n_1} + \frac{\sum \epsilon(0)}{n_0} + \frac{\sum \mu_1(x) - \mu_0(x)}{n} = 1.19$$

Assuming asymptotic normality:

$$(\hat{\tau}_{reg} - \tau) / \sqrt{\hat{V}_{superpop}} \sim N(0, 1)$$

a 95% confidence interval is constructed as:

$$\hat{\tau}_{reg} \pm 1.96 \sqrt{\hat{V}_{superpop}}$$

substituting values yields:

$$CI : [-6.15, -1.87]$$

The results suggest that controlling sex, gender, race, income, nerves medication use and life style, having at least a college-level education is associated with a statistically significant reduction in the change in smoking intensity between 1971 and 1982.

4.4 Sensitivity Analysis

To account for potential violations of the assumption of no unmeasured confounding in pair matching, we performed a Rosenbaum sensitivity analysis using the Wilcoxon signed rank test. We evaluated Γ at 1, 1.1, 1.15, 2, and 3. In Table 2, we find that at $\Gamma = 1.1$ the effect is statistically significant when testing at $\alpha = 0.05$, but at $\Gamma = 1.15$ the upper-bound p-value is 0.062, indicating that a hidden bias or unobserved covariate would need to increase the odds of exposure by more than a factor of $\Gamma = 1.1$ to explain away the observed association. Therefore, the findings are not robust to violations of conditional ignorability and are sensitive to small amounts of hidden bias.

To account for potential violations of the assumption of no unmeasured confounding in outcome regression, we could calculate the robustness value (Cinelli & Hazlett). The robustness value represents the “minimum strength of association (parameterized in terms of partial R^2) that omitted variables would need to have both with the treatment and with the outcome to change the estimated coefficient by a certain amount” [6]. A high robustness value would indicate the results are robust, while a low value would indicate high sensitivity.

In addition to assuming there were no unmeasured confounders, we assumed there was overlap to perform our analysis. To ensure the results are not driven from lack of overlap, a sensitivity analysis using the trimming method suggested by Crump et al. could be used. With this method, the data can be restricted to individuals with a propensity score in the range of [0.1,0.9]. This would ensure that the average treatment effect is estimated on data where both the treated and the control are well-represented [7].

Lastly, to address potential violations of the no interference assumption, we could consider using the randomization-based interference method by Rosenbaum [8]. This method provides

confidence intervals for treatment effect magnitudes without requiring the no interference assumption, allowing us to evaluate whether the magnitude of such interference would be sufficient to invalidate the estimated treatment effect.

4.5 Comparison

Both methods produced consistent and statistically significant estimates of the causal effect of higher education on changes in smoking intensity. The matching estimator yielded a mean difference of -3.53 cigarettes per day ($p = 0.019$, 95% CI: $[-6.50, -0.00]$), while the outcome regression estimator produced an ATE of -4.01 cigarettes per day (95% CI: $[-6.15, -1.87]$). These results suggest that individuals with at least a college-level education experienced a greater reduction in smoking intensity compared to those with lower education.

The two estimates are similar in magnitude, with outcome regression yielding a slightly larger effect. This difference is expected, as matching estimates the ATT within the matched sample, whereas outcome regression targets the ATE using the full sample and parametric assumptions.

While the results are consistent across methods, the sensitivity analysis indicates that the findings are not robust to even modest unmeasured confounding, as the effect becomes statistically insignificant when $\Gamma = 1.15$.

5 Conclusions and Limitations

This study examined the causal effect of higher education on changes in smoking intensity using NHEFS data. Both matching and outcome regression produced consistent and statistically significant estimates, indicating that higher education is associated with a reduction in smoking intensity.

However, the sensitivity analysis indicates that these findings are sensitive to even modest unmeasured confounding, preventing a strong causal interpretation. Therefore, while the results support a negative effect of education on smoking intensity, caution is warranted.

The analysis is further limited by potential unmeasured confounding and the reduced sample size after matching, which may restrict generalizability. Future work could improve robustness by incorporating additional confounders such as early-life socioeconomic factors. In addition, computing the robustness value [6] for the outcome regression estimate would help quantify sensitivity to unmeasured confounding. These findings highlight the importance of carefully assessing robustness when drawing causal conclusions from observational data.

Figures and Tables

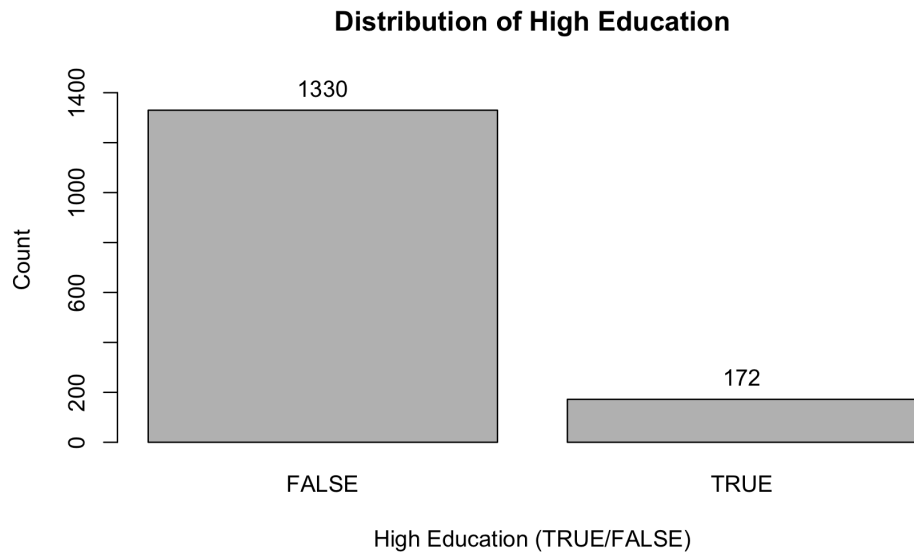


Figure 1: High Education (high_edu) Distribution

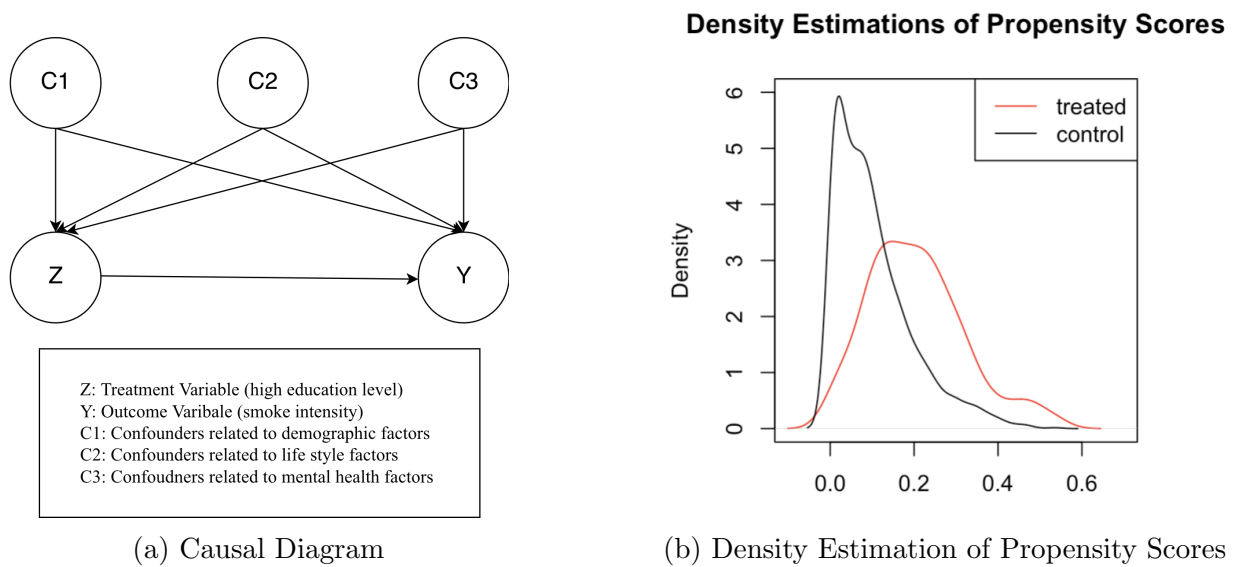


Figure 2: Diagnostic Plots

Covariate Balance Before and After Matching

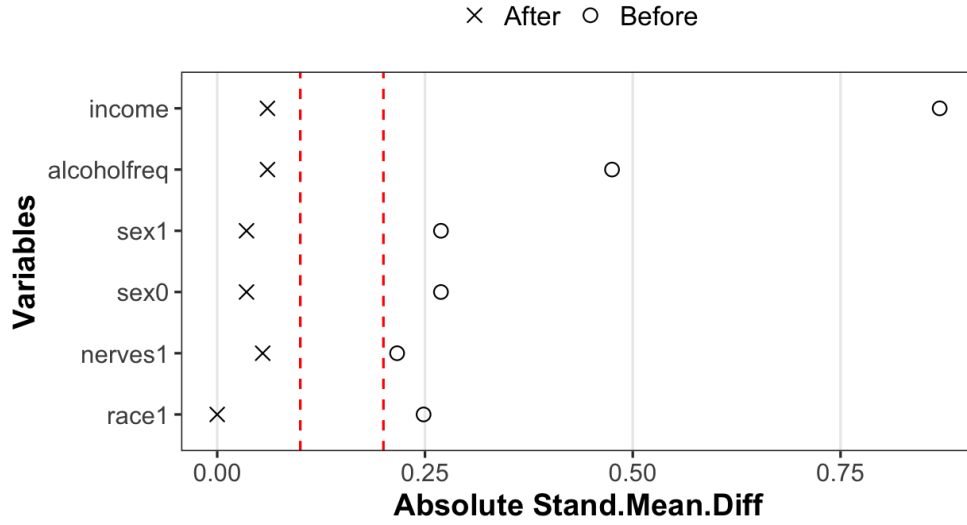


Figure 3: Love Plot

Table 1: Wilcoxon signed rank test results for paired comparison between y_1 and y_0

Statistic	Value
Test	Wilcoxon signed-rank test
V statistic	4799.5
p -value	0.019
Pseudo-median difference	-3.50
95% Confidence Interval	[-6.50, -0.00]

Table 2: Gamma and bounds on p -value from sensitivity analysis

Gamma	Lower Bound	Upper Bound
1.00	0.010	0.010
1.10	0.002	0.037
1.15	0.001	0.062
2.00	0.000	0.937
3.00	0.000	1.000

References

- [1] Centers for Disease Control and Prevention. Smoking tobacco use. <https://www.cdc.gov/tobacco/about/index.html>, 2024.
- [2] Vida Maralani. Understanding the links between education and smoking. <https://www.sciencedirect.com/science/article/pii/S0049089X14001057?via%3Dihub>, 2014.
- [3] Tian Geng, Caitlin E Smith, Chen Li, et al. Education and smoking behavior: Evidence from a twin study. *Scientific Reports*, 12:13841, 2022. URL: <https://www.nature.com/articles/s41598-022-17536-x>.
- [4] Nick Huntington-Klein and Malcolm Barrett. `nhefs_complete`: Cleaned `nhefs` data. https://rdr.io/cran/causaldata/man/nhefs_complete.html, 2024. R package documentation for the `causaldata` package.
- [5] National Center for Health Statistics. Nhanes i epidemiologic follow-up study (`nhefs`). <https://wwwn.cdc.gov/nchs/nhanes/nhefs/>, 2024. Accessed: 2026-04-18.
- [6] Carlos Cinelli and Chad Hazlett. Robustness value. https://carloscinelli.com/sensemakr/reference/robustness_value.html, 2020. Documentation for the `sensemakr` R package.
- [7] Richard K. Crump, V. Joseph Hotz, Guido W. Imbens, and Oscar A. Mitnik. Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1):187–199, 2009. URL: <https://academic.oup.com/biomet/article/96/1/187/235329>.
- [8] Paul R. Rosenbaum. Interference between units in randomized experiments. <https://www.tandfonline.com/doi/abs/10.1198/016214506000001112>, 2007.

Appendix

Appendix A: Variable Definitions

Variable	Definition
education	Amount of education by 1971, represented by the following: • 1: 8th grade or less • 2: high school dropout • 3: graduated high school • 4: college dropout • 5: completed college or more
high_edu	True if a level of education of at least college was completed; False otherwise
smkintensity82_71	Change in the number of cigarettes per day between 1971 and 1982
sex	0 if male; 1 if female
race	0 if white; 1 if black or otherwise
income	Total family income in 1971, represented by the following: • 11: < \$1,000 • 12: \$1,000–\$1,999 • 13: \$2,000–\$2,999 • 14: \$3,000–\$3,999 • 15: \$4,000–\$4,999 • 16: \$5,000–\$5,999 • 17: \$6,000–\$6,999 • 18: \$7,000–\$9,999 • 19: \$10,000–\$14,999 • 20: \$15,000–\$19,999 • 21: \$20,000–\$24,999 • 22: \$25,000+
alcoholfreq	Alcohol frequency consumption in 1971, represented by the following: • 0: Almost every day • 1: 2-3 times/week • 2: 1-4 times/month • 3: < 12 times/year • 4: No alcohol last year
nerves	1 if the individual uses nerves medication in 1971; 0 otherwise.

Appendix B: Data Summary Statistics

```

high_edu      smkintensity82_71    age      sex      race      income      alcoholfreq
Mode :logical  Min.   :-80.000  Min.   :25.0  0:731  0:1307  Min.   :11  Min.   :0.000
FALSE:1330     1st Qu.: -10.000  1st Qu.:33.0  1:771  1: 195  1st Qu.:17  1st Qu.:1.000
TRUE :172      Median :  0.000  Median :43.0                Median :19  Median :2.000
                Mean  : -4.525  Mean   :43.6                Mean  :18  Mean   :1.909
                3rd Qu.:  1.000  3rd Qu.:53.0                3rd Qu.:20  3rd Qu.:3.000
                Max.   : 50.000  Max.   :74.0                Max.   :22  Max.   :4.000

nerves
0:1288
1: 214

```

Appendix C: Data Sample

high_edu	smkintensity82_71	age	sex	race	income	alcoholfreq	nerves
TRUE	-10	28	0	0	19	0	1
TRUE	0	60	0	0	19	1	0
FALSE	-20	27	1	0	14	1	1
TRUE	-30	47	0	0	22	3	0
FALSE	0	53	0	0	18	0	1